Unit 4. Learning **[5 Hrs.]**

4.1. Concepts of machine learning
4.2. Rote learning
4.3. Learning by analogy
4.4. Inductive learning
4.5. Explanation based learning,
4.6. Supervised and unsupervised learning
4.7. Learning by evolution (genetic algorithm)

Learning in AI refers to the ability of machines to improve their performance or make better predictions over time, without being explicitly programmed for every task. This is achieved through various techniques like machine learning, which allows systems to learn from data and identify patterns.

Key Concepts in AI Learning:

- Machine Learning:

A subset of AI that focuses on enabling computers to learn from data without being explicitly programmed.

- Supervised Learning:

Training models with labeled data to make predictions or classifications.

- Unsupervised Learning:

Learning from unlabeled data to discover patterns and insights.

- Reinforcement Learning:

Training agents to make decisions in an environment to maximize rewards.

- Deep Learning:

A subfield of machine learning that uses neural networks to learn complex patterns from large amounts of data.

- Neural Networks:

Algorithms inspired by the structure and function of the human brain, used in deep learning.

- Natural Language Processing (NLP):

A field within AI focused on enabling computers to understand and generate human language.

How AI Learns:

- Data Analysis:

AI algorithms analyze vast amounts of data to identify patterns and relationships.

- Model Training:

AI models are trained on datasets, and their performance is evaluated and optimized.

- Decision Making:

Once trained, AI models can make decisions or predictions based on the learned patterns.

Types of AI Learning:

- Supervised Learning:

Uses labeled data to train models. For example, a model might be trained on images of cats and dogs, labeled as such, to learn to classify new images.

- Unsupervised Learning:

Uses unlabeled data to discover hidden patterns. For example, a model might be used to identify customer segments based on their purchasing behavior without knowing their individual identities.

- Reinforcement Learning:

Trains agents to make decisions in an environment to maximize rewards. For example, a robot might be trained to navigate a maze by receiving positive reinforcement for reaching the goal.

- Deep Learning:

Uses neural networks to learn complex patterns from large datasets. For example, a model might be trained on millions of images to learn to recognize objects.

Example:

An AI system could learn to identify different types of flowers from photographs. The system would be trained using a dataset of labeled images (e.g., "rose," "tulip," "sunflower"). As the system processes these images, it learns the features that distinguish each type of flower (e.g., petal shape, color, size). Once trained, the system can accurately classify new, unseen flower photographs.

4.1 Concepts of machine learning

Machine learning is a subset of artificial intelligence that automatically enables a machine or system to learn and improve from experience. Instead of explicit programming, machine learning uses algorithms to analyze large amounts of data, learn from the insights, and then make informed decisions.

It enables machines to learn from data without explicit programming, allowing them to improve performance on specific tasks over time. In essence, ML provides the algorithms and models that allow computers to analyze data, identify patterns, and make predictions or decisions.

Machine learning (ML) is a specialized subset of artificial intelligence (AI) that focuses on developing algorithms and statistical models that enable computers to learn from data and

improve their performance automatically without being explicitly programmed. While AI represents the broad vision of creating machines that simulate human cognitive functions, machine learning serves as the primary data-driven method used to achieve that intelligence.

Core Hierarchy of AI

To understand where machine learning fits, it helps to view the ecosystem as nested layers:

- **Artificial Intelligence (AI)**: The overarching field encompassing any system that mimics human-like thinking, perception, reasoning, or problem-solving. This includes rule-based logic systems, expert systems, and robotics.
- **Machine Learning (ML)**: A distinct domain inside AI where computers rely on data inputs rather than manually written rules to figure out tasks on their own.
- **Deep Learning (DL)**: A specific, specialized subfield of machine learning that utilizes complex, multi-layered artificial neural network to process highly sophisticated, non-linear relationships like audio or high-definition imagery.

Here's a more detailed explanation:

- **ML as a Subfield of AI:**

AI encompasses the general concept of enabling machines to perform tasks that typically require human intelligence, while ML focuses on how to make machines learn and adapt.

- **Learning from Data:**

ML algorithms are trained on large datasets, allowing them to learn relationships and patterns within the data.

- **Improving Performance:**

As ML models are exposed to more data, they can refine their predictions and decision-making abilities, leading to improved performance.

Types of ML:

There are several types of ML, including:

- Supervised learning: Machines learn from labeled data to make predictions.
- Unsupervised learning: Machines learn from unlabeled data to discover patterns and structures.
- Reinforcement learning: Machines learn through trial and error, receiving rewards or penalties for their actions.

Applications of ML:

ML is used in a wide range of applications, including:

- Image recognition: Identifying objects and features in images.
- Natural language processing: Enabling computers to understand and generate human language.
- Recommendation systems: Suggesting items or content based on user preferences.
- Fraud detection: Identifying suspicious transactions or patterns.
- Autonomous vehicles: Enabling self-driving cars.

4.2 Rote learning: =>

Rote Learning is a learning process where the AI system stores and reproduces data or patterns without the ability to generalize and apply knowledge effectively.

In the context of AI, rote learning, also known as rote memorization, refers to a form of learning where an AI system stores information without necessarily understanding or applying it. This can be seen as a simple memorization process where the system learns facts or patterns through repetition or by comparing new inputs to a storied history of previous inputs and outputs.

The detailed description is as given as below:

- **Memorization:**

Rote learning in AI focuses on memorizing data or patterns without necessarily grasping the underlying concepts or relationships.

- **Storage:**

The system stores new information for later retrieval and use.

- **Simple Learning:**

It can be considered a basic form of learning, often used in systems that need to quickly recognize or classify information.

- **Example:**

An AI system might recognize faces by storing features from images and comparing them to a database of known faces. If a match is found, the person is recognized based on the stored information.

In essence:

Rote learning in AI is a method where the system learns by simply storing information and retrieving it when needed, similar to how a human memorizes facts or formulas through repetition.

4.3 Learning by analogy

Learning by analogy in AI involves an AI system acquiring new knowledge by transferring understanding from a similar known entity or situation to a new, related entity or situation. This process helps AI systems generalize and apply existing knowledge to novel scenarios, improving problem-solving and decision-making capabilities.

Here's a more detailed explanation:

1. What is Learning by Analogy?

- Learning by analogy is a cognitive process where an AI system identifies similarities between two entities or situations (the "source" and the "target") and then transfers knowledge, relationships, or solutions from the source to the target.
- For example, an AI system might learn to play chess by analogizing it to other board games, like checkers, and transferring concepts of strategy and move planning.

- It's a powerful mechanism for exploiting past experience, allowing AI to leverage knowledge from similar situations to solve new problems.

2. How it works:

- Identify Similarity:

The AI system first needs to identify the relevant similarities between the source and target entities or situations.

- Transfer Knowledge/Solutions:

Once a similarity is established, the AI system transfers knowledge, relationships, or solutions from the source to the target.

- Generalization:

This process allows the AI system to generalize and apply the transferred knowledge to other related situations, even if they haven't been explicitly trained on them.

3. Examples and Applications:

- Problem-solving:

An AI system might learn to solve a new puzzle by analogizing it to a previously solved puzzle.

- Planning:

An AI system could use past experiences to plan routes or manage resources by analogizing them to similar past scenarios.

- Understanding:

An AI system could learn new concepts by analogizing them to concepts it already understands.

- Self-driving cars:

Self-driving cars can use analogies to previous experiences to handle novel road conditions.

4. Benefits and Challenges:

- Benefits:

Analogical reasoning allows AI systems to learn more efficiently, adapt to new situations, and solve problems creatively.

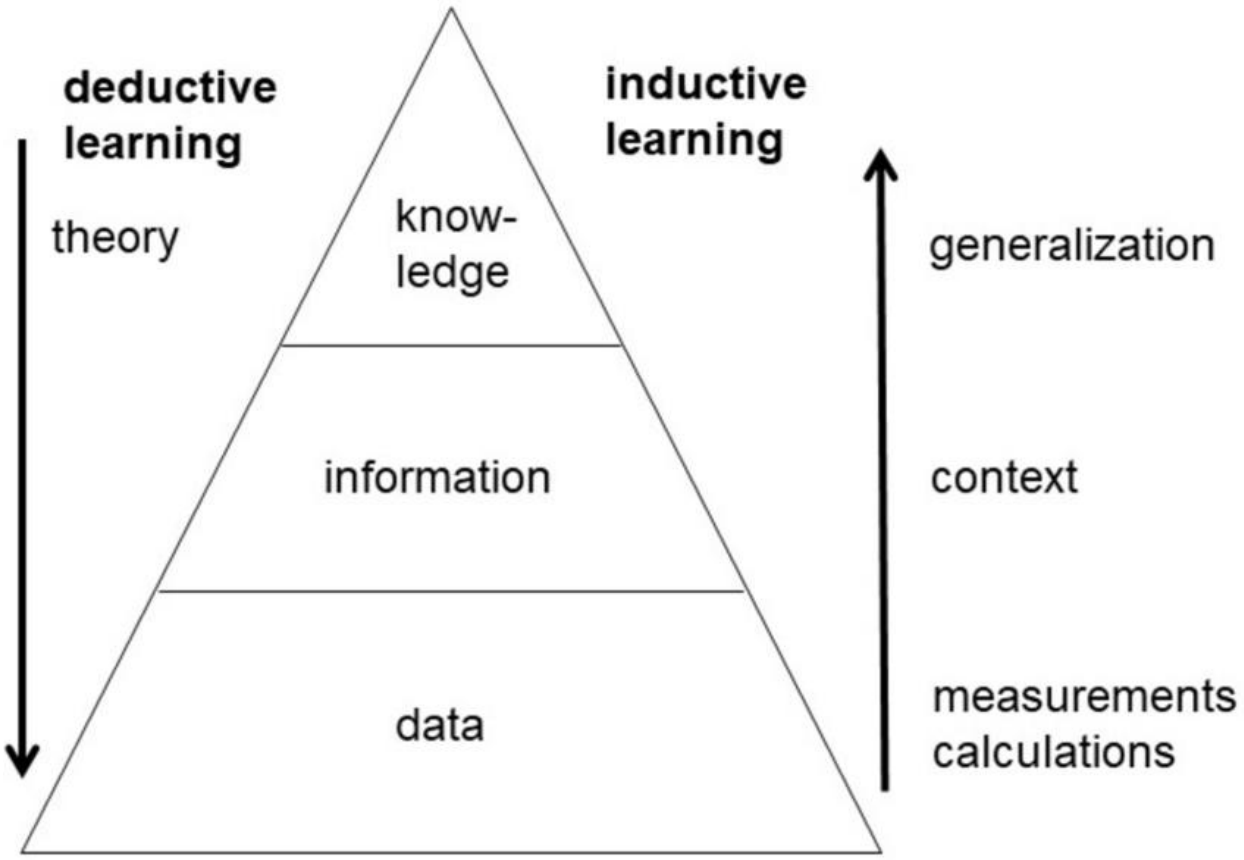
- Challenges:

AI systems need a lot of background knowledge, common sense, and creativity to effectively use analogies. They also need to be able to represent, retrieve, compare, and evaluate different sources of information.

5. Importance in AI:

- Learning by analogy is a fundamental mechanism of thought that is essential for AI to achieve its potential.
- It helps AI systems to generalize, adapt, and solve problems in situations that are not explicitly trained on.
- It is crucial for developing AI systems that can reason, learn, and make decisions in complex, real-world scenarios

4.4 Inductive learning:



Inductive Learning (bottom to top in the pyramid):
It means starting from data and building up to knowledge or theory.

1. Start with data

→ Raw facts or numbers (e.g., temperatures, prices, survey results).

2. Turn data into information

→ Add context or meaning (e.g., "The temperature increased over 5 days").

3. From information to knowledge

→ Find patterns or rules (e.g., "When the sky is clear, temperatures rise").

4. Create generalizations or theories

→ Use that knowledge to make conclusions or predictions about new situations.

Inductive learning, also known as inductive reasoning or inductive inference, is a type of learning that involves generalizing from specific instances or examples to make broader generalizations or predictions. It is a fundamental approach in machine learning and is closely related to the concept of inductive bias.

Inductive learning makes a generalization from specific observations and facts and inductive learning uses bottom-up approach

Inductive learning in AI involves a model learning from examples to make predictions or classifications about new, unseen data. This is achieved by observing patterns and generalizing rules from specific instances. For example, a model trained on images of cats and dogs can learn to recognize these animals in new images, even if it hasn't seen those specific images before.
Here are some examples of inductive learning in AI:

- **Spam email classification:**

A model learns to identify spam emails based on features like the number of links, keywords, and sender's address.

- **House price prediction:**

A model learns the relationship between house features (bedrooms, size, location) and price, allowing it to predict the price of new houses based on their features.

- **Image recognition:**

A model learns to identify objects in images (e.g., cats, dogs, birds) by analyzing visual patterns in labeled datasets.

- **Sentiment analysis:**

A model learns to predict the sentiment (positive, negative, or neutral) of text by analyzing words and phrases in reviews or social media posts.

- **Medical diagnosis:**

A model learns to diagnose diseases by analyzing patient data and medical history.

- **Voice recognition:**

A model learns to recognize speech patterns and translate spoken words into text.

- **Recommendation systems:**

A model learns user preferences and recommends products or content based on past behavior.

- **Credit risk assessment:**

A model learns to assess the creditworthiness of individuals by analyzing their financial data and other relevant information.

- **Predictive maintenance in manufacturing:**

A model learns to predict when equipment is likely to fail based on historical data.

4.5 Explanation based learning

Explanation-Based Learning (EBL) in Artificial Intelligence is a technique where a system learns by understanding and generalizing from specific examples, leveraging domain knowledge to infer rules and concepts. It's a method that contrasts with traditional inductive learning, which relies on large datasets to identify patterns.

Key aspects of EBL:

- Leverages Domain Knowledge:

EBL utilizes prior knowledge about a specific domain to explain why a training example belongs to a particular concept or class.

- Focuses on Understanding:

Instead of simply identifying statistical patterns, EBL aims to understand the underlying principles that govern the example.

- Generalizes from Single Examples:

EBL can learn general rules from a single example, by analyzing the example and using domain knowledge to infer the relevant properties.

- Improves Learning Efficiency:

By leveraging domain knowledge, EBL can accelerate learning, improve the accuracy of learned concepts, and increase confidence in the learned knowledge.

- Examples:

A system learning to diagnose car engine failures might use domain knowledge (e.g., how engines work) to explain why an engine fails to start. By analyzing the diagnostic information, it can generalize this explanation to form a rule for future diagnostic scenarios. Another example is a system learning to play chess, which could use the rules of chess (domain knowledge) to explain why a particular chess position leads to a forced loss of the black queen.

How EBL works:

1. Input Example: The system is presented with a specific training example.
2. Use of Domain Knowledge: The system utilizes pre-existing domain knowledge to explain the example.
3. Explanation Generation: The system generates an explanation for why the example is an instance of the concept.
4. Generalization: The system generalizes the explanation to form a rule or concept.
5. Learning Outcome: The system has learned a new rule or concept that can be applied to future examples.

Advantages of EBL:

- Efficiency:

EBL can learn from a small number of examples, especially when combined with strong domain knowledge.

- Accuracy:

EBL can achieve higher accuracy in learning compared to inductive methods, especially when the domain knowledge is accurate and relevant.

- Explainability:

EBL provides explanations for the learned rules, making it easier to understand and debug the system's reasoning.

- Adaptability:

EBL can be adapted to different domains by incorporating different domain knowledge.

Applications of EBL:

- Diagnostics: EBL can be used to learn diagnostic rules in various domains, such as medical diagnosis, computer system troubleshooting, and industrial process monitoring.
- Problem Solving: EBL can be used to learn problem-solving strategies and heuristics.

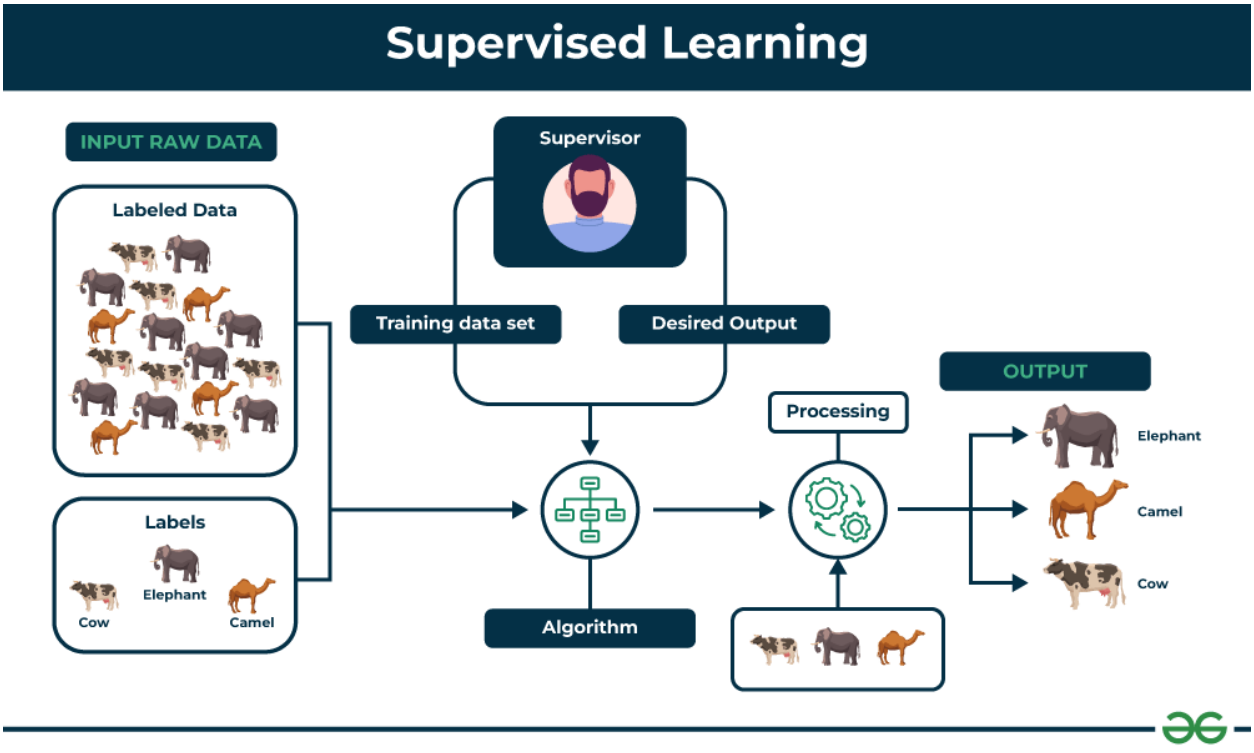
- Knowledge Acquisition: EBL can be used to acquire new knowledge and rules from examples.

4.6 Supervised and unsupervised learning

Supervised and unsupervised learning are two fundamental approaches in machine learning. Supervised learning uses labeled data to train models that can make predictions on new, unseen data, while unsupervised learning discovers patterns and structures within unlabeled data.

Supervised Learning:

- **Training Data:** Uses labeled datasets where both inputs and outputs are known.
- **Goal:** To learn a relationship between inputs and outputs to make accurate predictions.
- **Examples:** Spam detection, medical diagnosis, handwriting recognition.
- **Algorithms:** Regression (e.g., linear regression), classification (e.g., logistic regression), decision trees, support vector machines.
- **Benefits:** Can be more accurate than unsupervised learning for specific tasks when labeled data is available.
- **Limitations:** Requires labeled data, which can be expensive and time-consuming to obtain.



Unsupervised Learning:

- **Training Data:**

  Uses unlabeled, or "raw" data where the desired output is not explicitly given.

- **Goal:**

  To discover hidden patterns, structures, and relationships within the data.

- **Examples:**

  Customer segmentation, anomaly detection, recommendation systems, image segmentation.

  Algorithms:

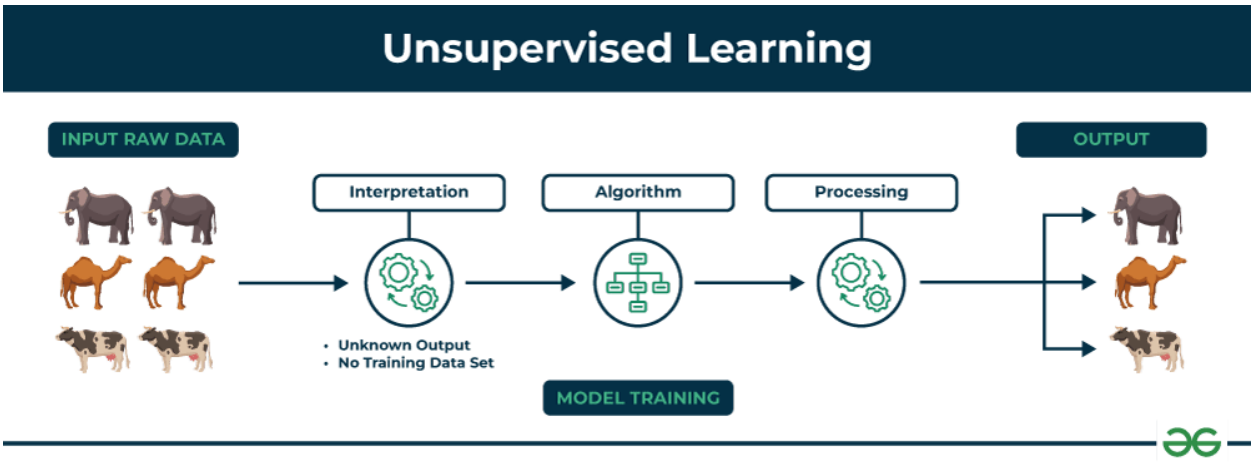
  Clustering (e.g., k-means, hierarchical clustering), dimensionality reduction (e.g., PCA), association rule mining.

- **Benefits:**

  Can be used on large datasets, can discover hidden patterns that supervised learning might miss, and can be used for tasks where labeled data is not available.

- **Limitations:**

  May lack clear accuracy metrics and can be more prone to subjective interpretation.



Key Differences:

Feature	Supervised Learning	Unsupervised Learning
Data	Labeled	Unlabeled
Goal	Prediction	Discovery of patterns

Examples	Spam detection, medical diagnosis	Customer segmentation, anomaly detection
Algorithms	Regression, classification	Clustering, dimensionality reduction

Supervised vs Unsupervised Machine Learning

Parameters	Supervised machine learning	Unsupervised machine learning
Input Data	They are trained on labeled data.	They are trained on unlabeled data.
Computational Complexity	Simpler method	Computationally complex
Accuracy	Highly accurate	Less accurate
No. of classes	No. of classes is known	No. of classes is not known
Data Analysis	Uses offline analysis	Uses real-time analysis of data
Algorithms used	Linear and Logistics regression, KNN Random forest, multi-class classification, decision tree, Support Vector Machine, Neural Network etc.	K-Means clustering, Hierarchical clustering, Apriori algorithm etc.
Output	Desired output is given.	Desired output is not given.
Training data	Use training data to infer model.	No training data is used.

Parameters	Supervised machine learning	Unsupervised machine learning
Complex model	It is not possible to learn larger and more complex models with supervised learning.	It is possible to learn larger and more complex models with unsupervised learning.
Model	We can test our model.	We can not test our model.
Supervision	Supervised learning needs supervision to train the model.	Unsupervised learning does not need any supervision to train the model.
Classification	Divided into two types: 1. Regression 2. Classification	Divided into two types: 1. Clustering 2. Association
Feedback	It has feedback mechanism.	It has no feedback mechanism.
Time Consumption	It's more time consuming.	It's less time consuming.
Example	Optical character recognition.	Find a face in an image.

4.7 Learning by evolution (genetic algorithm): =>

Learning by evolution, also known as evolutionary algorithms, is a type of artificial intelligence (AI) approach that draws inspiration from the natural process of evolution. Instead of explicitly programming algorithms, these methods use evolutionary processes like mutation, crossover, and selection to generate and refine solutions to problems.

Learning by Evolution in AI

- **Inspired by Nature**
  AI uses ideas from **natural evolution** (like how animals evolve over time).
  Just like stronger animals survive and have babies, AI keeps and improves the best solutions to a problem.
- **Working with Many Solutions**
  Instead of working with just **one solution**, it works with a **group of solutions** (called a population).
  Each one is a different guess or approach to solving the problem.
- **How It Improves Over Time**
  The AI improves the solutions using three main steps:

1. **Mutation**: Makes small random changes to a solution to try something new.
2. **Crossover**: Mixes parts of two good solutions to create a new one.
3. **Selection**: Picks the best ones from each group to keep and improve in the next round.

- **Where It's Used**
  Evolution-based AI can be used for:

1. **Tuning settings** in other AI models (called hyperparameters).
2. **Choosing important data features**, making models simpler and faster.
3. **Creating programs automatically** to solve tasks.
4. **Solving tough optimization problems** in areas like engineering and finance.

- **Why It's Useful**

1. **Robust**: Works even when data is messy or unclear.
2. **Scalable**: Can handle big, complex problems.
3. **Flexible**: Can adjust when the problem changes over time.

A genetic algorithm (GA) is an optimization algorithm inspired by the principles of natural selection and biological evolution. It uses a population of candidate solutions, called individuals or chromosomes, and iteratively modifies them to find the best solution to a problem. The basic steps in a GA include creating an initial population, evaluating the fitness of each individual, selecting parents for reproduction, performing crossover (mixing genetic material), and mutation (introducing random changes).

Key Concepts:

- **Population:** A set of candidate solutions to a problem.
- **Individual/Chromosome:** A representation of a potential solution, encoded as a string of parameters.
- **Fitness Function:** A function that evaluates the quality of a solution, determining how "good" it is.
- **Selection:** Choosing individuals for reproduction based on their fitness.

- **Crossover:** A genetic operator that combines parts of two parent solutions to create offspring.
- **Mutation:** A genetic operator that introduces random changes to an individual's parameters.

How it Works:

1. **Initialization:** A random population of individuals is created.
2. **Evaluation:** The fitness of each individual is calculated using a fitness function.
3. **Selection:** Individuals with higher fitness values are selected as parents for reproduction.
4. **Crossover:** Parents' genetic material is recombined to create offspring.
5. **Mutation:** Random changes are introduced to the offspring's parameters.
6. **Iteration:** The process of evaluation, selection, crossover, and mutation is repeated for multiple generations, until a satisfactory solution is found or a maximum number of generations is reached.

Applications:

1. **Optimization:** Finding the best solution to a problem, such as minimizing a cost function or maximizing a profit.
2. **Machine Learning:** Training models, such as neural networks, by optimizing their parameters.
3. **Feature Selection:** Identifying the most relevant features in a dataset.
4. **Evolutionary Programming:** Developing algorithms that evolve their own code.

Advantages:

1. **Global Optimization:** GAs are capable of finding the global optimum solution, even in complex search spaces.
2. **Robustness:** GAs are relatively robust to noisy or incomplete data.
3. **Parallelism:** GAs can be easily parallelized, allowing for faster computation.

Limitations:

1. **Parameter Tuning:** Finding the right parameters for a GA can be challenging.
2. **Computational Cost:** GAs can be computationally expensive, especially for large problems.
3. **Local Optima:** While GAs strive for global optima, they can sometimes get stuck in local optima.

In essence, a genetic algorithm mimics the evolutionary process to find optimal solutions to problems by iteratively improving a population of candidate solutions through selection, crossover, and mutation.

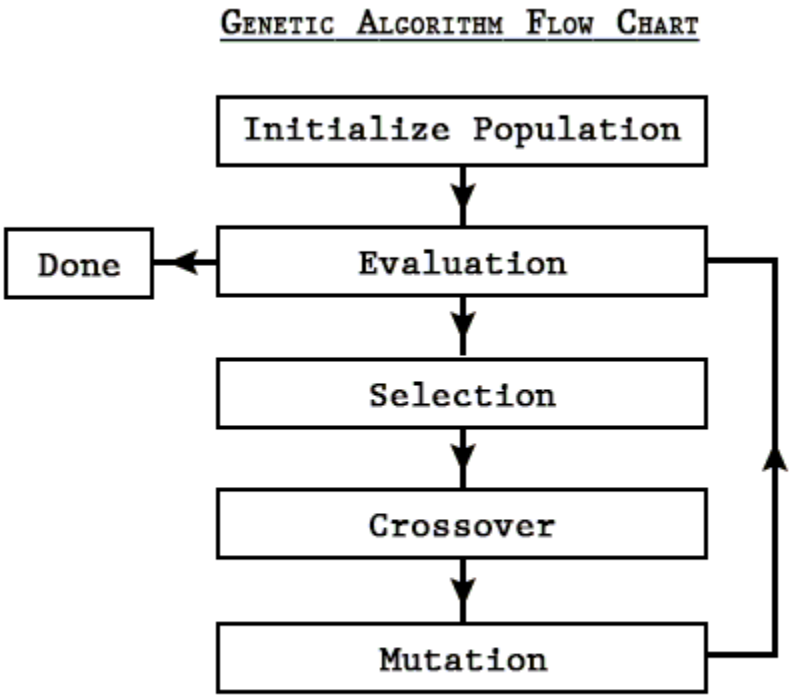


FIGURE 2

In simple words:

A **Genetic Algorithm** is a search and optimization technique based on the principles of **natural selection** and **genetics**. It is one of the most popular ways to implement **learning by evolution** in AI.

Basic Steps of a Genetic Algorithm

Here's how the algorithm works, step-by-step:

1. **Initialization**
   - Create a random population of potential solutions (called chromosomes or individuals).
2. **Evaluation**
   - Measure how good each solution is using a **fitness function** (like a score or accuracy).

3. **Selection**
    - Choose the best solutions to be parents (those with higher fitness).
4. **Crossover (Recombination)**
    - Combine parts of two parents to make new solutions (offspring).
5. **Mutation**
    - Apply small random changes to some offspring to explore new possibilities.
6. **Replacement**
    - Form a new generation by replacing some or all of the old population with the new solutions.
7. **Repeat**
    - Go back to step 2 and repeat until the solution is good enough or a set number of generations is reached.

Example Use Case

Let's say you want to **optimize a neural network's hyperparameters**. GA can try different combinations of learning rates, number of layers, and activation functions — and evolve toward the best setup.

Common Applications

- Function optimization
- Machine learning model tuning
- Feature selection
- Game strategy learning
- Automated design (e.g., circuits, robots)

Pseudocode of Genetic Algorithm

```
1. Initialize population with random solutions
2. Evaluate fitness of each solution
3. While stopping condition not met:
   a. Select parents based on fitness
   b. Apply crossover to create offspring
   c. Apply mutation to offspring
   d. Evaluate fitness of offspring
   e. Replace some or all of the population
4. Return the best solution found
```